

Executive Analysis

The reasoning behind the 2-slide Tour-e proposal

Maruti Suzuki XCELerate 2026 · E-Track · Round 1 supporting document

Summary

India's ride-service fleet is priced on one number: **cost per kilometre**. Platforms compete on fares and driver take-home, so the vehicle that wins is the one that carries the most paying passengers at the lowest ₹/km with the least downtime. Our answer is **Tour-e**: a compact electric MPV of about 4.40 m (Ertiga-class footprint), seating 5+2, with a 45 kWh usable LFP battery, about 300 km of real-world range and a kerb-side powered sliding door. It is sold to fleets with **Battery-as-a-Service (BaaS)** and run on **Maruti FleetOS**.

₹7.9/km

Tour-e + BaaS, 5-yr TCO
(owner-driver, 60k km/yr)

₹10.1/km

Ertiga Tour M CNG
same duty cycle

+₹10.9k

extra owner-driver
income per month

₹1.5-1.8 L

fleet saving per car
per year (80k km)

This document explains the reasoning behind each choice on our two slides: the segments we target, why the body is an MPV, why 5+2 seats, why a 45 kWh LFP pack, why the strategy is BEV-first but not BEV-only, and why a long-range variant is needed. Detailed evidence sits in the accompanying files:

02_TCO_Model.xlsx (live model), **03_Market_Research**, **04_Technical_Specifications**, **05_FleetOS_Architecture** and **06_Sources_and_References**.

1 · The problem

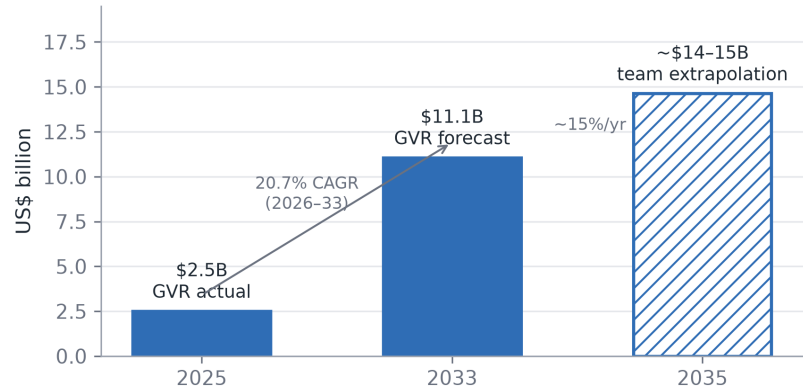
The E-Track brief asks for the ideal ride-service vehicle for India in 2035, one that serves intra-city and inter-city demand while maximising passenger comfort, driver income and fleet profitability. Three facts shape the answer:

- **The market is large and growing fast.** India's ride-hailing market was US\$2.51B in 2025 and is forecast to reach US\$11.06B by 2033, a 20.7% CAGR (Grand View Research). Extrapolating at a slower ~15%/yr gives ~US\$14-15B by 2035 (our estimate).
- **Fleet economics are running-cost driven.** A cab covers 60,000-80,000 km a year, so energy and maintenance outweigh purchase price. CEEW (2025) finds electric taxis the cheapest option to own, with CNG next.
- **Today's fleet is CNG sedans and MPVs.** Maruti's Tour range has passed 5 lakh sales and the Ertiga sold 190,974 units in FY25, India's No.1 MPV. These buyers already trust the Ertiga shape and Maruti's service network, and Tour-e electrifies that shape rather than asking the market to adopt a new one.

2 · Where the volume and the margin sit

Intra-city trips were 60.75% of India's taxi trips in 2025, and airport transfers are the fastest-growing trip purpose at a 19.14% CAGR (Mordor Intelligence). We therefore design for the city as the volume base and for airport and inter-city trips as the margin pools.

Use case	Role	What it needs	Tour-e mission
Intra-city on-demand	VOLUME	Lowest ₹/km, easy entry, 200-250 km/day uptime	City, 5 seats
Daily commute / corporate	STEADY DEMAND	Seat count per trip, predictable routes	Shuttle, 7-seat pooled
Airport transfers	MARGIN	Boot space, comfort, reliability	5 seats + ~400 L boot
Inter-city	MARGIN	Range, highway charging	Tour-e Long; CNG off-corridor
Shared / premium / accessible	NICHE MARGIN	Space, rear AC, step-free access	7-seat · premium trim · WAV



India ride-hailing market size. 2025 and 2033: Grand View Research. 2035: team extrapolation at ~15%/yr after 2033.

3 · Why an MPV rather than a sedan or SUV

Sedans dominate cabs today because they balance price, boot space and CNG-tank packaging: the Dzire took 56% of sedan sales in August 2026. An EV removes the tank-versus-boot trade-off and places the battery under a flat floor, so the best body is the one that turns that floor into cabin space and easy entry. That points to a tall, compact MPV.

Criterion	Compact sedan	Compact SUV	Compact MPV (Tour-e)	Van (Eco-type)
Seats per trip	4	4-5	5 + 2 fold-flat	5-7
Airport luggage with full seats	Medium	Medium	High (~400 L in 5-seat mode)	High
Packaging a floor battery	Raises floor, cramps rear	Good, but heavy and costly	Good: tall body keeps headroom	Good, poor crash safety
Entry / exit, kerb-side	Low seat, hinged door	High step	Sliding door, ≥720 mm opening	Sliding door
Energy use (weight, drag)	Lowest	Higher	Moderate	Moderate
Market familiarity in fleets	High (Dzire / Tour S)	Low	High (Ertiga / Tour M)	Medium
Fit for 2035 ride-service	Good for city only	Premium niche	Best all-round	Low-cost niche

Precedent: Toyota's purpose-built JPN Taxi is 4.40 m long with a 720 mm powered sliding door on the kerb side and a 401 L boot. Japan drives on the left, as India does, so the same door layout suits Indian kerbs.

4 · Why 5+2 seating

- **Most trips carry 1-3 passengers.** Five seats is the default, which keeps a ~400 L boot for airport luggage.
- **A fold-flat third row adds 2 seats** for pooled office shuttles, families and group airport runs. The same car can serve a corporate shuttle in the morning and on-demand city trips in the afternoon, which raises paid kilometres per day.
- **One body, several products.** The same package supports a wheelchair-accessible variant (ramp and wheelchair bay in place of the second-row kerb-side seat) and a 7-seat retail twin that adds volume and lowers unit cost.

5 · Why ~45 kWh of LFP

The battery is the most expensive part of the car (36% of our 2028 cost stack), so we size it to the duty cycle rather than to a range headline.

- **Duty day:** owner-drivers cover ~200 km/day (60,000 km over 300 days) and double-shift fleets ~250 km/day. At 150 Wh/km real-world consumption with AC on, 45 kWh usable gives ~300 km, leaving a 100 km buffer for owner-drivers and 50 km for fleets.
- **End of life:** at 70% state of health the pack still gives ~210 km, enough for a full owner-driver day, so the car keeps earning across a 4 lakh km, 5-year design life (~1,330–1,450 equivalent full cycles).
- **Cost:** every extra 10 kWh adds ~₹0.8 L of cost at ₹8,000/kWh plus weight that raises ₹/km. A 60 kWh pack is kept for the inter-city variant only.
- **Chemistry:** LFP uses no nickel or cobalt, tolerates heat better than NMC and is the cheapest pack chemistry: BNEF's 2025 survey puts average LFP packs at US\$81/kWh and the China average for all packs at US\$84/kWh.
- **Charging:** 10–80% of a 45 kWh pack is 31.5 kWh. Doing that in ≤30 min needs ~63 kW on average, so a 100–120 kW peak is enough. A 150 kW system would add cost without cutting charging time much at this pack size.

6 · Why BEV-first, not BEV-only

	BEV (Tour-e)	Strong hybrid	Plug-in hybrid	CNG (Tour M)	Bio-CNG (CBG)
Energy cost per km	₹1.87	~₹4.5–5 (est.)	~₹3 (est.)	₹4.25	≈ CNG
Maintenance per km	₹0.55	Medium	High (two drivetrains)	₹0.95	≈ CNG
GST on the vehicle	5%	40% (>4 m)	40% (>4 m)	40% (>4 m)	40% (>4 m)
Range / refuel	~300 km; ≤30 min DC	700 km+; 5 min	Long; 5 min	~250 km; 5 min	~250 km; 5 min
Infrastructure today	22,753 public chargers + depots	Everywhere	Everywhere	Growing to 18,000 stations by 2030	Few plants
Tailpipe emissions	Zero	Low	Low / zero in EV mode	Low	Low; renewable fuel
Role in our plan	Primary	Not chosen	Not chosen	Bridge	Bridge fuel

Energy cost per km for BEV and CNG from our model (₹85/kg CNG at 20 km/kg; 150 Wh/km at a blended ₹11.2/kWh). Hybrid figures are team estimates. GST: GST Council, effective 22 Sep 2025. Chargers: Ministry of Heavy Industries, Rajya Sabha, Mar 2026. CNG stations: MoPNG target.

The 35-point GST gap (5% vs 40%) and the ~₹2.4/km energy saving make the BEV the lowest-cost choice wherever depot charging exists. Charging access is still uneven, so the plan does not assume BEVs everywhere:

- **1 · BEV** for high-utilisation urban fleets with depot charging (the volume and airport segments).

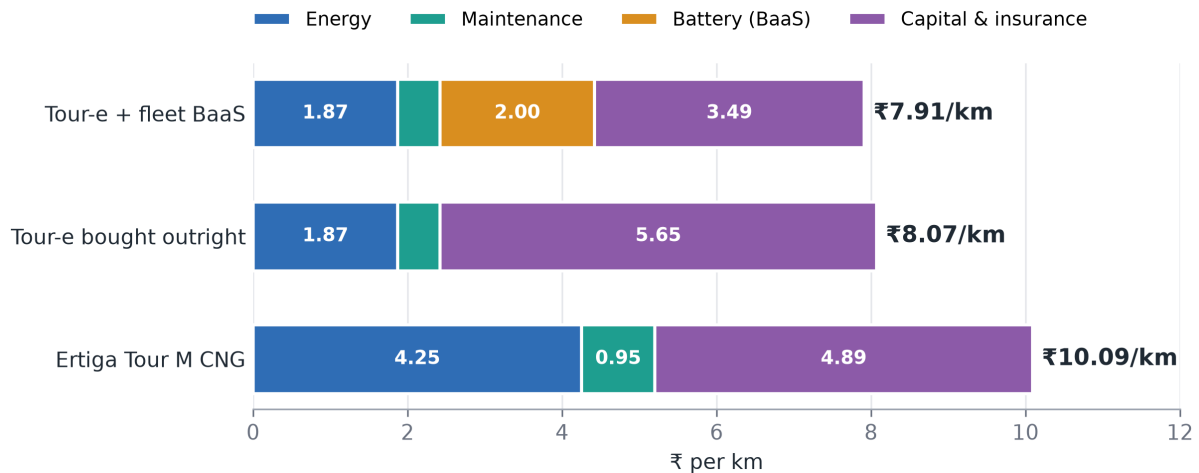
- **2 · CNG / CBG bridge:** in cities or corridors where charging is constrained, operators keep using the **existing Ertiga Tour M**. It needs no new engineering, and Maruti's ₹561 crore investment in four biogas (CBG) plants shows it can supply renewable gas.
- **3 · Tour-e Long** (~60 kWh) for inter-city corridors with highway fast charging.

We rejected a CNG version of the Tour-e body. Packaging a CNG tank and engine into a platform designed around a floor battery would add cost and risk for a bridge that the existing Ertiga Tour M already provides.

7 · Why Tour-e Long is needed

Inter-city trips are a margin segment, and typical routes such as Delhi-Jaipur or Chennai-Bengaluru run 280–350 km one way. The standard 45 kWh car would need a stop on every such trip. A ~60 kWh version gives ~400 km at 150 Wh/km, which covers most one-way corridor trips on a single charge and is topped up at highway DC chargers before the return leg. It shares the body, eAxle and electronics with the standard car, so it adds volume to the same line rather than a separate programme. Off the charging corridors, the CNG Tour M stays the right tool.

8 · The economics in one view

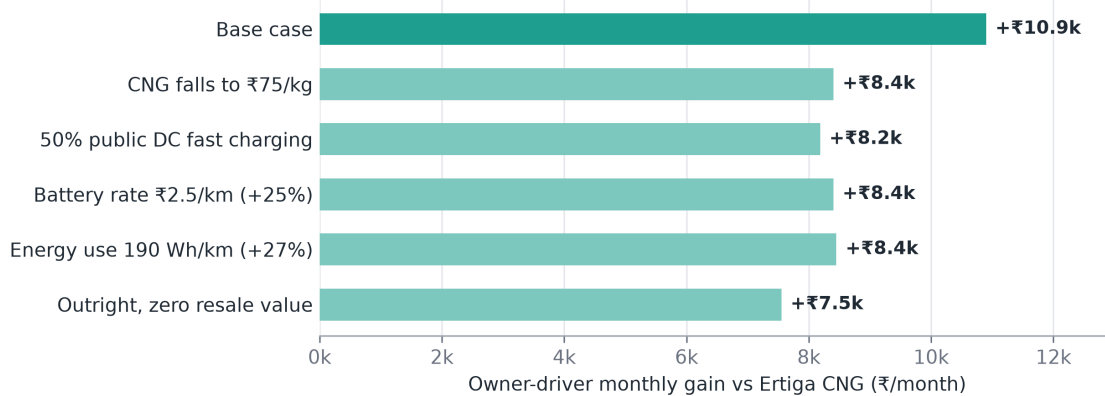


5-year cost per km at 60,000 km/yr (owner-driver). Team model, 2028 launch prices, 2026 ₹. Source: 02_TCO_Model.xlsx, TCO sheet.

	Ertiga Tour M CNG	Tour-e + BaaS	Tour-e outright
Upfront price	₹10.74 L	₹7.9 L + ₹2.0/km battery	₹12.4 L
Cost per km @ 60k km/yr (owner-driver)	₹10.09	₹7.91	₹8.07
Owner-driver gain vs CNG per month	-	+₹10,900	+₹10,100
Cost per km @ 80k km/yr (fleet)	₹8.86	₹7.03	₹6.66
Fleet saving per car per year	-	₹1.5 L	₹1.8 L
Payback of extra purchase price	-	Immediate (cheaper upfront)	~9 months

Why the battery rate is ₹2.0/km. A BaaS provider buying the 48 kWh pack at ₹4.15 L, financing it at 11% over five years with a 25% residual value, a 10% risk reserve and an 8% margin breaks even at ₹1.89/km on a 60,000 km/yr minimum contract. Maruti's retail e VITARA BaaS is ₹3.99/km because it is priced on ~22,000 km/yr; a fleet spreads the same pack over ~2.7× the distance.

Stress tests. The Tour-e stays ahead if CNG falls to ₹75/kg (+₹8.4k/month), if half of all charging is at public DC chargers (₹8.45/km), if the battery rate is 25% higher, or if real-world energy use is 27% worse than target. CNG would have to fall to ~₹41/kg to erase the gap.



Owner-driver monthly gain vs Ertiga CNG under stress tests. Source: 02_TCO_Model.xlsx, Sensitivity sheet.

9 · Key assumptions

Assumption	Value	Basis
Annual km: owner-driver / fleet	60,000 / 80,000	~200 / ~250 km per working day
Tour-e energy use (at battery)	150 Wh/km	Design target, AC on; 45 kWh ÷ 150 Wh/km = 300 km
Charging mix and tariffs	80% depot @ ₹9 · 20% public DC @ ₹20/kWh	State EV tariffs; CPO pricing 2026
CNG price and efficiency	₹85/kg · 20 km/kg real	Delhi ₹86.98/kg (Sep 2026); claimed 26.11 km/kg
Maintenance: Tour-e / CNG	₹0.55 / ₹0.95 per km	No oil, filters or clutch; regen braking
Finance · insurance · horizon	10.5% · 3%/yr · 5 years	Commercial vehicle loans
Residual value at year 5	15% (CNG, EV outright) · 20% (BaaS body)	High-mileage taxi resale
Battery pack cost 2028 / 2035	₹8,000 / ₹6,500 per kWh	≈ BNEF 2025 China average \$84/kWh at ₹95/\$
Price build-up	Cost × 1.08 × 1.04 × 1.05	OEM margin, dealer margin, 5% GST
Excluded (conservative)	Road-tax / permit waivers, state incentives	Would further favour the EV

10 · Risks and mitigation

Risk	Mitigation
Heat and battery fade	Liquid-cooled LFP pack, plugged-in pre-cooling, BMS charge-rate limits in heat; BaaS risk reserve carries replacement risk
Charging gaps outside metros	Depot hubs with CPOs, OMCs and DISCOMs; CNG/CBG bridge with the Ertiga Tour M; Tour-e Long on corridors
Import dependence (cells, magnets)	Dual-sourced cells (FinDreams and ACC-PLI makers); reduced rare-earth motor; pack, BMS and thermal localised first
Residual value uncertainty	SoH battery passport, buy-back programme, battery owned by the BaaS provider
Fleet operator failure (the BluSmart case)	Asset-light model: many operators and owner-drivers, NBFC-funded BaaS, no single-operator exposure

11 · Conclusions

- **Vehicle:** a ~4.40 m electric MPV with 5+2 seats and a kerb-side sliding door electrifies the body type Indian fleets already buy and adds step-free, luggage-friendly packaging.
- **Powertrain:** BEV-first with a 45 kWh LFP pack sized to the duty day, the Ertiga Tour M on CNG/CBG as the bridge, and a ~60 kWh Tour-e Long for inter-city routes.
- **Economics:** ₹7.9/km against ₹10.1/km for CNG means +₹10.9k a month for an owner-driver and ₹1.5–1.8 L a year per fleet car, and the advantage survives every stress test we ran.
- **Digital layer:** FleetOS turns vehicle data into battery-aware dispatch, charge scheduling and predictive maintenance, which protects uptime and residual value.
- **Scale:** prove on e VITARA fleets (2026–28), launch Tour-e with airport and corporate fleets first (2028–31), and scale to at least 1 lakh units a year by 2035, with the price falling from ₹12.4 L to ₹10.6 L.